

NetPulse OpenVPN

Diagnostics & RouterOS Onboarding Cheat Sheet

Current setup: dedicated NetPulse management VPN on TCP 1195

Read this first

This guide is for the dedicated NetPulse RouterOS management VPN only. The separate generic OpenVPN service on TCP 1194 uses a different certificate authority and must not be stopped, edited, or migrated without explicit verification.

1. Setup at a glance

Values below describe the working arrangement established on August 26, 2026. Verify the public IP and DNS before using them in a new environment.

NetPulse service	openvpn-server@netpulse
NetPulse config	/etc/openvpn/server/netpulse.conf
NetPulse endpoint	netpulse.co.ke:1195/TCP
Other VPN	TCP 1194 — separate service; leave it alone
VPN subnet	10.8.0.0/24 (MGR received 10.8.0.2)
Client identity	CN=netpulse-mgr
RouterOS cipher	RouterOS 6: aes128 RouterOS 7: aes128-cbc
Server cipher	AES-128-CBC + data-ciphers-fallback AES-128-CBC

2. Golden safety rules

- Repair only the marked NetPulse file and unit: /etc/openvpn/server/netpulse.conf and openvpn-server@netpulse.
- Do not stop openvpn@server or edit /etc/openvpn/server.conf unless its ownership has been explicitly confirmed.
- A different CA fingerprint means the other service is a separate VPN. Treat it as unrelated.
- Back up the NetPulse config before every manual edit.
- Do not regenerate the NetPulse CA during routine troubleshooting; CA rotation invalidates previously issued router certificates.
- Do not change RADIUS or customer-traffic settings while the VPN itself is unhealthy.

Triage workflow

An open port, a certificate handshake, and a fully established tunnel are different checkpoints.

3. Diagnosis order

- 1. Service: `openvpn-server@netpulse` is active and stable.
- 2. Listener: TCP 1195 is owned by the NetPulse OpenVPN process.
- 3. Reachability: an external check reports TCP 1195 open.
- 4. Certificate: logs show VERIFY OK for NetPulse CA and the router CN.
- 5. Cipher: no data-channel cipher negotiation errors.
- 6. Tunnel: Peer Connection Initiated and a 10.8.0.x address appear.
- 7. Router: RouterOS reports `netpulse-vpn` connected/running.

4. Core Linux commands

Run these on the Ubuntu server.

```
sudo systemctl status openvpn-server@netpulse --no-pager -l
sudo systemctl is-active openvpn-server@netpulse
sudo journalctl -u openvpn-server@netpulse -n 80 --no-pager
sudo journalctl -u openvpn-server@netpulse -f
```

Listener, services, and process ownership

```
sudo ss -ltnp | grep -E ':(1194|1195)\b' || true
sudo systemctl list-units --type=service 'openvpn*' --all
sudo ps -ef | grep '[o]penvpn'
```

Inspect only the dedicated configuration

```
sudo grep -nE '^(port|proto|dev|ca|cert|key|server|cipher|data-ciphers|data-ciphers-fallback|auth|
username-as-common-name|auth-user-pass-verify)' \
/etc/openvpn/server/netpulse.conf
```

Important

TCP 1194 being open does not prove NetPulse is healthy. In this setup, 1194 belongs to the separate generic service; NetPulse must be verified on 1195.

Failure signatures

Use the message, not just the symptom, to select the repair.

5. Common log messages

Options error: username-as-common-name

The dedicated config still contains a username-only option. Remove username-as-common-name and auth-user-pass-verify for certificate-only RouterOS authentication.

No common cipher

The router supports AES-128-CBC but the server advertises only modern GCM/ChaCha ciphers. Align the dedicated server to AES-128-CBC.

VERIFY ERROR: self-signed certificate / unknown CA

Usually an old router profile is presenting a certificate from a different CA. Do not regenerate the current CA blindly; identify which client attempt is current.

TCP connection established but no Peer Connection Initiated

Network reachability works, but TLS, certificate, authentication, or cipher negotiation is still failing.

Port open but wrong process owns it

An open port is not proof that NetPulse is running. Confirm the PID, command line, service, and config path.

6. Safe CA comparison for a duplicate service

This check prints fingerprints and does not stop anything.

```
sudo bash <<'CHECK'
set -euo pipefail
N=/etc/openvpn/server/netpulse.conf
L=/etc/openvpn/server.conf
get() { awk -v k="$1" ' $1 == k { print $2; exit } ' "$2"; }
NCA="$(get ca "$N")"
LCA="$(get ca "$L")"
echo "NetPulse CA: $(openssl x509 -in "$NCA" -noout -fingerprint -sha256)"
echo "Other CA:    $(openssl x509 -in "$LCA" -noout -fingerprint -sha256)"
CHECK
```

Interpretation

If the fingerprints differ, do not stop the generic service automatically. Move NetPulse to a free port instead, then update the portal and router provisioning endpoint to match.

Dedicated NetPulse repair

The following repair leaves the separate TCP 1194 VPN untouched.

7. Repair the port, authentication mode, and cipher

```
sudo bash <<'FIX'
set -euo pipefail
CONFIG=/etc/openvpn/server/netpulse.conf
BACKUP="$CONFIG.netpulse-final-$(date +%Y%m%d%H%M%S).bak"
cp -p "$CONFIG" "$BACKUP"
sed -i -E \
  -e 's/^[[[:space:]]*port[[[:space:]]+[0-9]+/port 1195/' \
  -e 's/^[[[:space:]]*proto[[[:space:]]+tcp([[[:space:]]*)$)/proto tcp-server\1/' \
  -e '/^[[[:space:]]*username-as-common-name([[[:space:]]|$)/Id' \
  -e '/^[[[:space:]]*auth-user-pass-verify([[[:space:]]|$)/Id' \
  -e 's/^[[[:space:]]*cipher([[[:space:]]+).*/cipher AES-128-CBC/I' \
  -e 's/^[[[:space:]]*data-ciphers([[[:space:]]+).*/data-ciphers AES-128-CBC/I' \
  -e 's/^[[[:space:]]*data-ciphers-fallback([[[:space:]]+).*/data-ciphers-fallback AES-128-CBC/I' \
  "$CONFIG"
systemctl restart openvpn-server@netpulse
sleep 3
systemctl is-active openvpn-server@netpulse
ss -ltnp | grep -E ':(1194|1195)\b' || true
journalctl -u openvpn-server@netpulse -n 25 --no-pager
echo "Backup: $BACKUP"
FIX
```

If cipher lines do not already exist in the dedicated file, add:

```
cipher AES-128-CBC
data-ciphers AES-128-CBC
data-ciphers-fallback AES-128-CBC
auth SHA1
```

8. Do not stop the other VPN

The generic process using /etc/openvpn/server.conf and TCP 1194 may be a separate product or customer service. A different CA fingerprint confirms that it is not the NetPulse instance. Leave it running.

Port rule

NetPulse uses TCP 1195 because TCP 1194 is occupied by the separate service. The portal setting, server config, upstream firewall/NAT rule, and router profile must all agree on 1195.

Reachability and RouterOS

After the server is healthy, verify the path and reconnect the router.

9. External reachability

Run from a different network or use an external TCP checker. Testing locally on the server only proves a local listener.

```
nc -vz -w5 netpulse.co.ke 1195
getent ahostsv4 netpulse.co.ke
```

Firewall note

This setup used iptables-nft/netfilter-persistent rather than UFW. Do not add UFW rules blindly. Confirm the upstream firewall, NAT, or provider security rule allows TCP 1195 to this server.

10. Reconnect the RouterOS VPN client

```
/interface ovpn-client disable [find name="netpulse-vpn"]
:delay 2s
/interface ovpn-client enable [find name="netpulse-vpn"]
:delay 10s
/interface ovpn-client monitor [find name="netpulse-vpn"] once
```

If only the port changed and the existing certificate is current

```
/interface ovpn-client set [find name="netpulse-vpn"] port=1195
/interface ovpn-client disable [find name="netpulse-vpn"]
:delay 2s
/interface ovpn-client enable [find name="netpulse-vpn"]
/interface ovpn-client monitor [find name="netpulse-vpn"] once
```

11. Successful server log

```
VERIFY OK: depth=1 ... CN=NetPulse CA
VERIFY OK: depth=0, CN=netpulse-mgr
Peer Connection Initiated
MULTI: primary virtual IP ... 10.8.0.2
```

What not to misread

Older attempts may still show unknown CA. The current attempt is the one that shows VERIFY OK for the current NetPulse CA and router CN, followed by Peer Connection Initiated.

Onboard many routers

The shared provisioning path is reusable, but certificates remain unique per router.

12. Multi-router checklist

- Set Infrastructure !' VPN Port to 1195 and leave Protocol as TCP.
- Keep the actual NetPulse server config and public firewall/NAT rule on TCP 1195.
- Keep the NetPulse CA stable. Do not regenerate the CA during routine onboarding.
- Generate one client certificate for each router. Never copy one client private key to multiple routers.
- Download and run that router's .rsc setup script. It receives the saved port and RouterOS-version-compatible cipher.
- Confirm the VPN first. Only then rely on the remaining RADIUS, bridge, PPPoE, hotspot, and callback setup.
- If the port or CA changes later, reprovision all affected routers or update them deliberately.

13. What is shared versus per router

Shared

Server address, VPN port, TCP protocol, VPN subnet, and server cipher policy come from the global VPN configuration.

Per router

Router name, provisioning token, client certificate/key, client common name, and assigned VPN IP are unique to each router.

RouterOS cipher names

RouterOS 6 uses aes128. RouterOS 7 uses aes128-cbc. The generated script selects the name based on the reported RouterOS version.

Existing routers

If a router was configured for 1194, update its port or rerun its provisioning script after the portal is saved as 1195. Certificate regeneration is not required just because the port changed.

Rollback and final verification

Use the backup created before the manual edit if a change causes a new failure.

14. Restore only the dedicated NetPulse config

```
ls -lt /etc/openvpn/server/netpulse.conf.netpulse-*.bak
sudo cp -p /etc/openvpn/server/netpulse.conf.netpulse-<timestamp>.bak \
/etc/openvpn/server/netpulse.conf
sudo systemctl restart openvpn-server@netpulse
sudo systemctl status openvpn-server@netpulse --no-pager -l
```

15. Success criteria

- openvpn-server@netpulse is active and stable.
- TCP 1195 is owned by the dedicated NetPulse OpenVPN process.
- The server logs show VERIFY OK for the NetPulse CA and the router CN.
- The logs show Peer Connection Initiated with no cipher negotiation error.
- RouterOS reports netpulse-vpn connected/running and receives a 10.8.0.x address.
- Only after the above: proceed with customer-service provisioning and traffic validation.

16. Stop and investigate if

- The CA comparison shows different CAs and someone proposes stopping the generic service.
- TCP 1194 is assumed to be NetPulse merely because it is open.
- The NetPulse unit is not the process listening on 1195.
- The router certificate CN is not the expected router identity.
- The server shows certificate success but still reports no common cipher.

Runbook owner note

Keep the portal VPN port, installed server config, upstream firewall/NAT, and router profiles synchronized. That alignment is the key requirement for reliable onboarding of many routers.